www.tntextbooks.in

a) Clock Speed

Every microprocessor has an **internal clock** that regulates the speed at which it executes instructions. The speed at which the microprocessor executes instructions is called the **clock speed**. Clock speed is measured in MHz (Mega Hertz) or in GHz (Giga Hertz).

b) Instruction Set

A command which is given to a computer to perform an operation on data is called an **instruction**. Basic set of machine level instructions that a microprocessor is designed to execute is called as an **instruction set**. This instruction set carries out the following types of operations:

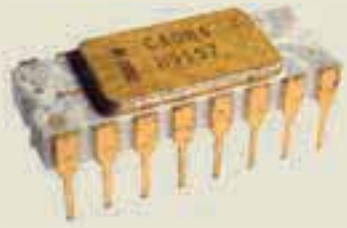
- Data transfer
- Arithmetic operations
- Logical operations
- Control flow
- Input/output

c) Word Size

- The number of bits that can be processed by a processor in a single instruction is called its word size. **Word size** determines the amount of RAM that can be accessed by a microprocessor at one time and the total number of pins on the microprocessor. Total number of input and output pins in turn determines the architecture of the microprocessor.

The first commercial microprocessor, Intel 4004 is a 4 bit processor. It has 4 input pins and 4 output pins. Number of output pins is always equal to the number of input pins. It can process 4 bits at a time. So it is called as a 4 bit processor.

Intel 4004



Produced	From late 1971 to 1981
Manufacturer	Intel Inc.
Clock Speed	740 kHZ
Size	10 Micrometer (µm)
Transistors	2300
Data width	4 bits
Package	16 pin

Intel Inc. released many microprocessors like Intel 8085 which is an 8 bit processor, Intel 8086 which is a 16 bit processor and so on. Currently most of the microprocessors use 32 bit or 64 bit architecture.

3.3 Data communication between CPU and memory

The Central Processing Unit(CPU) has a Memory Data Register (MDR) and a Memory Address Register (MAR). The Memory Data Register (MDR) keeps the data which is transferred between the Memory and the CPU. The Program